



RAI 811: ROBOT MECHANICS & CONTROL



Profile

Dr. Karam Dad Kallu

Assistant Professor

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Work Experience

Assistant Professor (16th Aug 2019-Till date)

Department of Robotics and Artificial Intelligence (R&AI), School of Mechanical and Manufacturing Engineering (SMME), National University of Sciences and Technology (NUST), Islamabad, Pakistan

Post-doctoral Researcher (15th Oct 2023- 14th Jan 2025)

MeRIC-Lab (Medical Robotics & Intelligent Control Laboratory), School of Mechanical Engineering, Chonnam National University, Gwangju, South Korea

Visiting Lecturer (2013-2015)

Islamic International University, Islamabad, Pakistan



PhD (2019)

Mechanical Engineering (Intelligent Control and Automation Systems), Pusan National University, Busan, South Korea

Thesis Topic: Tele-operated Bilateral Control of Hydraulic Manipulator Using Sliding Mode Control with Sliding Perturbation Observer

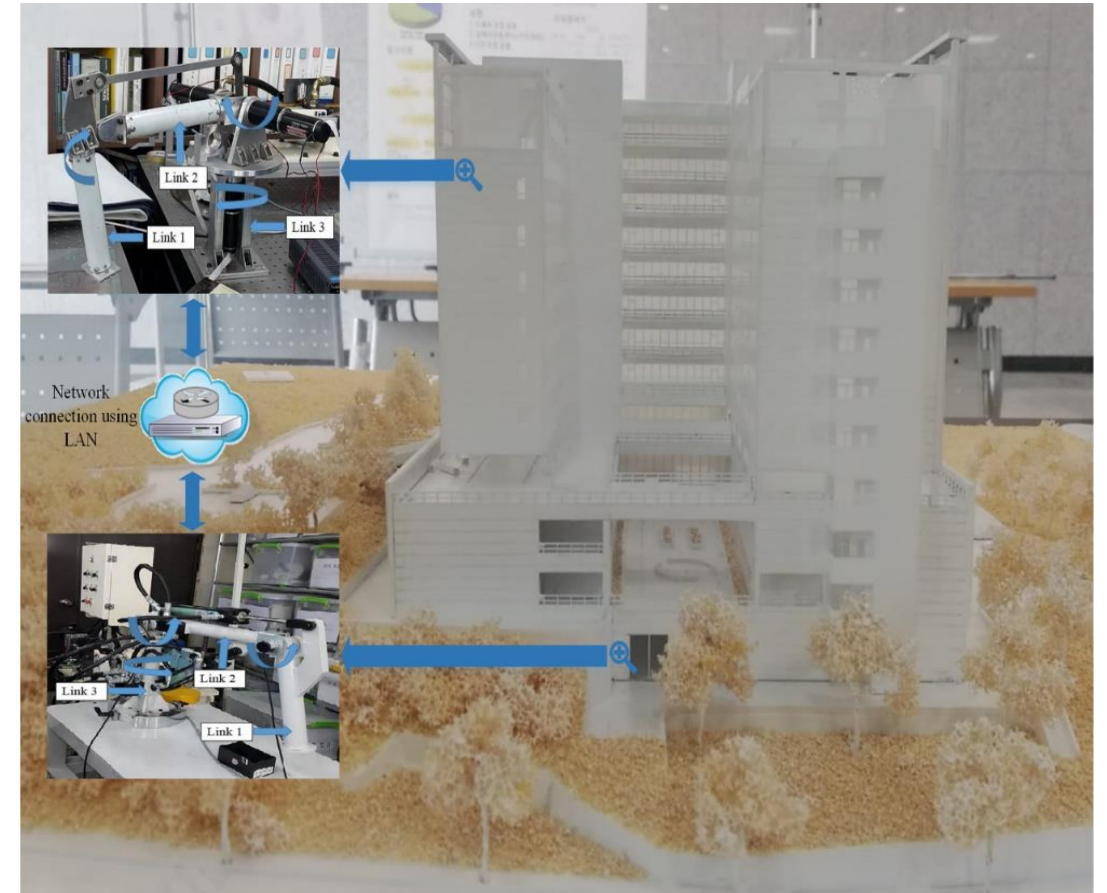
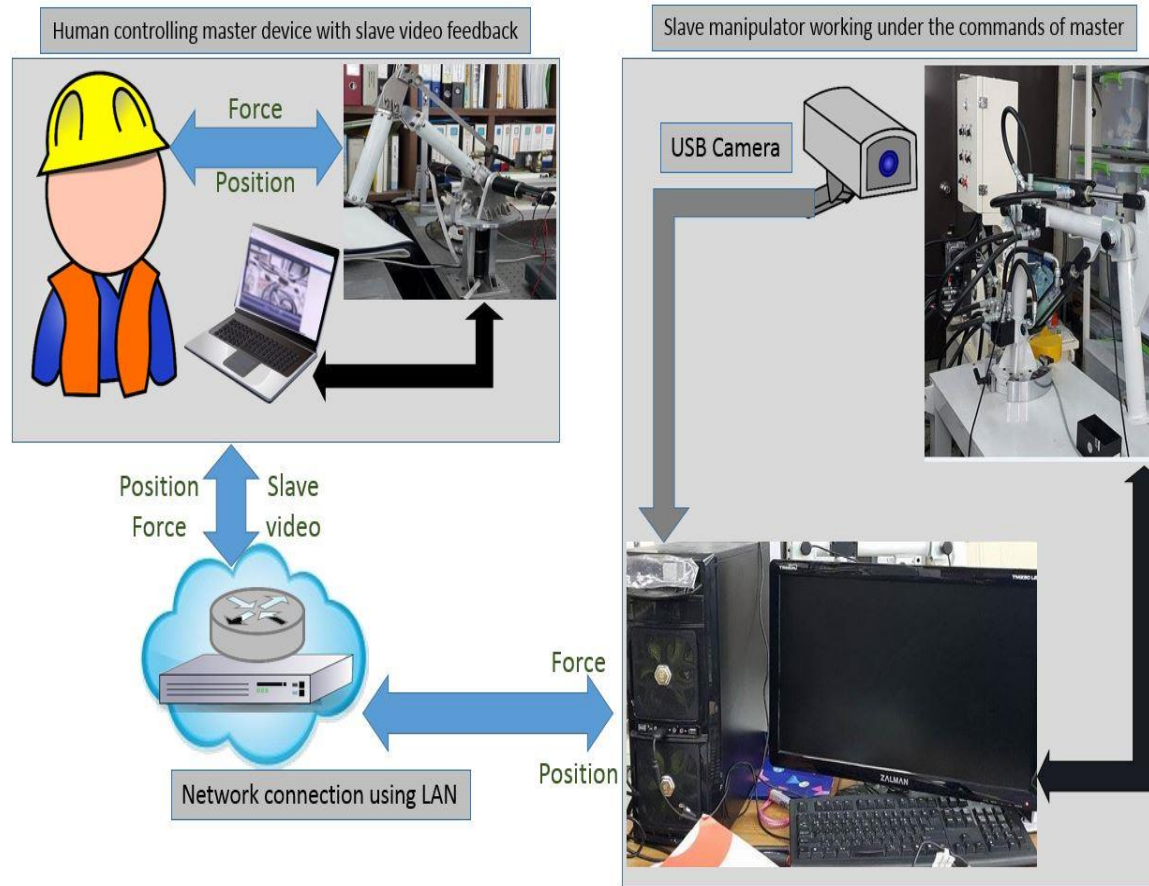
Supervisor: Prof. Dr. Min Cheol Lee

Key Areas: AI (RL), Non-linear Control, Robust Control, Adaptive Control, Bilateral Control, Human Robot Interaction, Robot Manipulator and System Identification

Medical Robotics, Surgical Robotics, Haptics, Tele-operation Force Sensing Module for Medical Application, Surgical Assistant Robot

Research

“To demonstrate internet based tele-operating bilateral control technique, which has force feedback without force sensor and minimum or without time delay problem for tele operating dismantle”



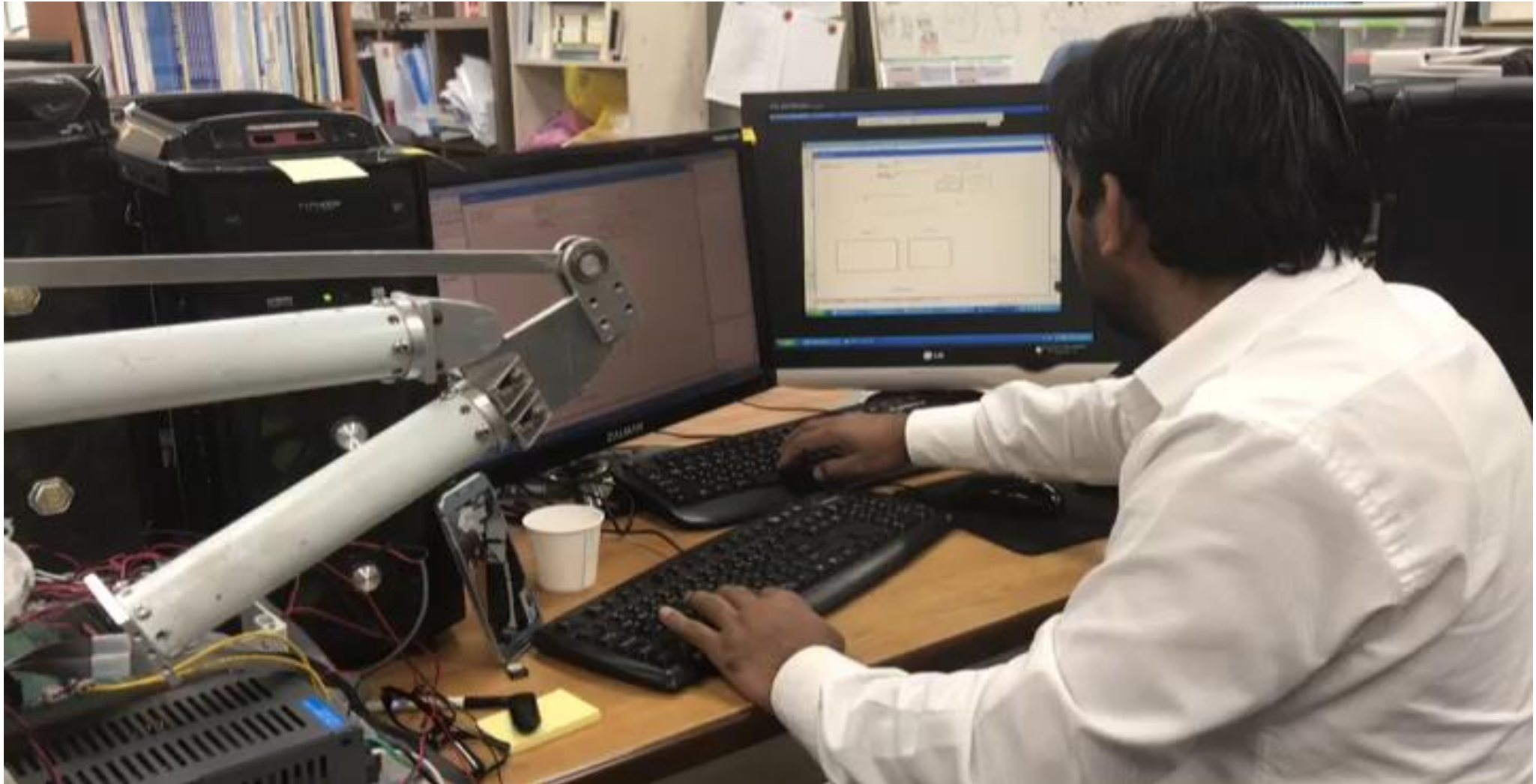


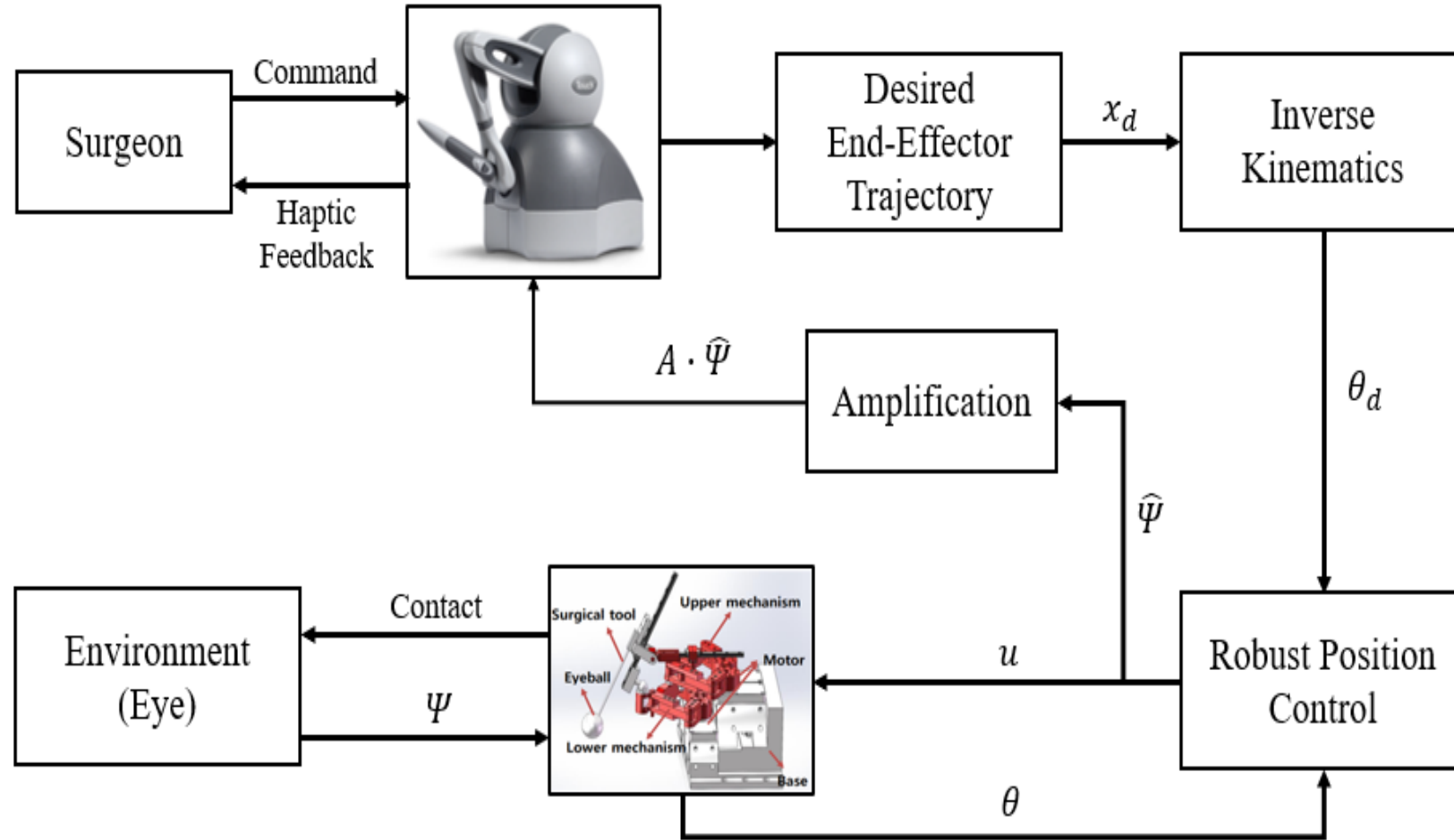
Research



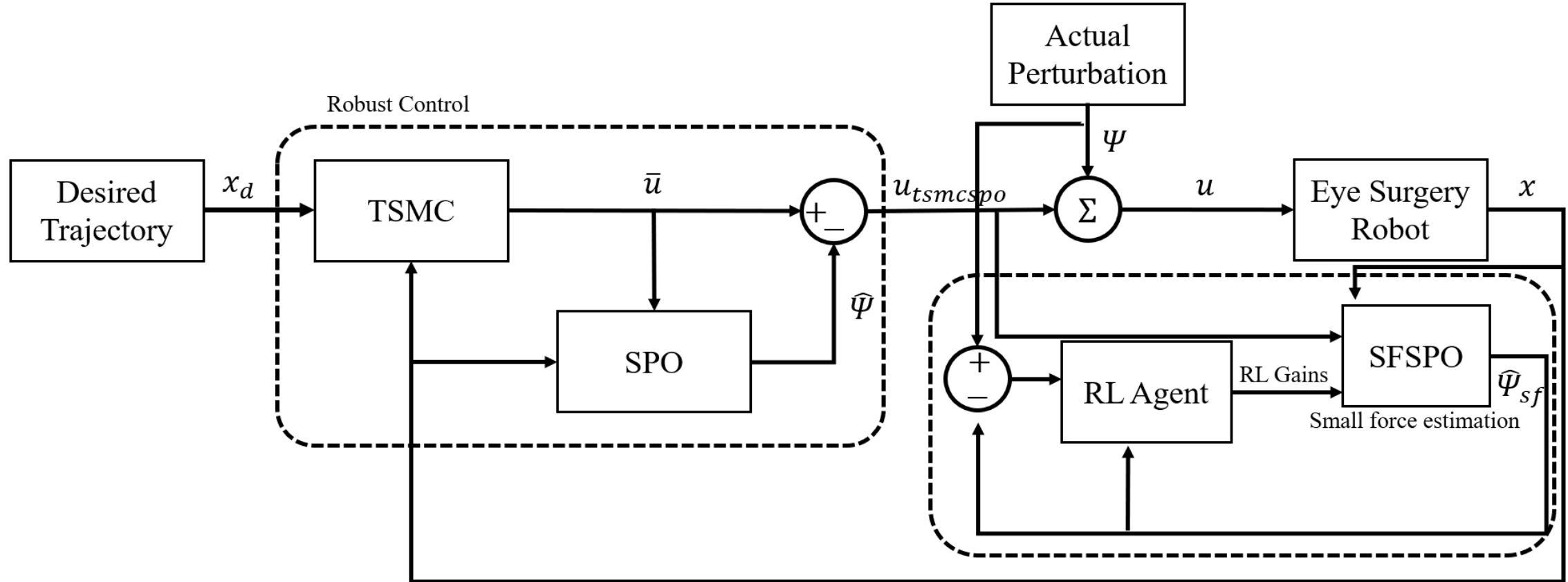


Research





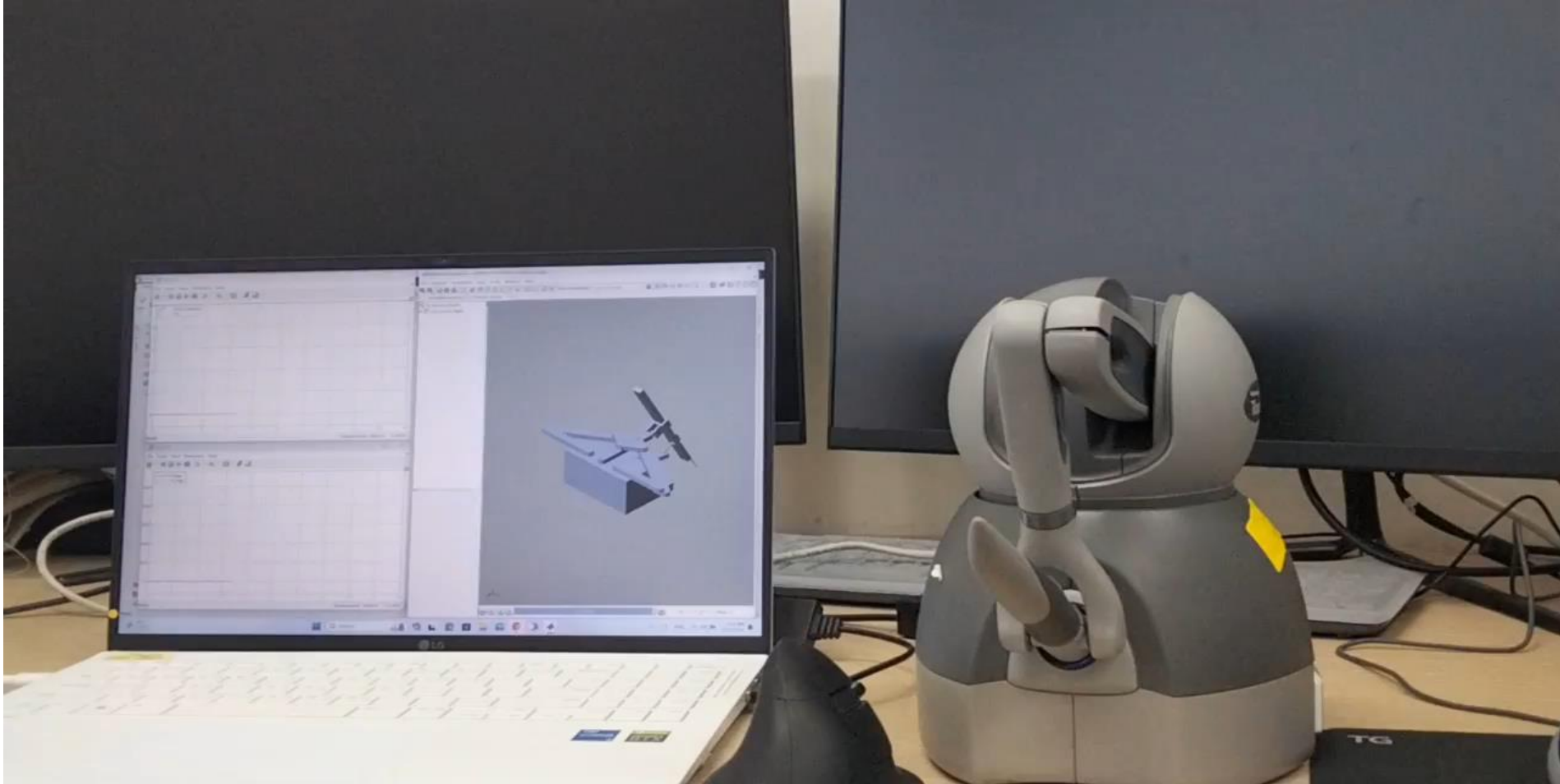




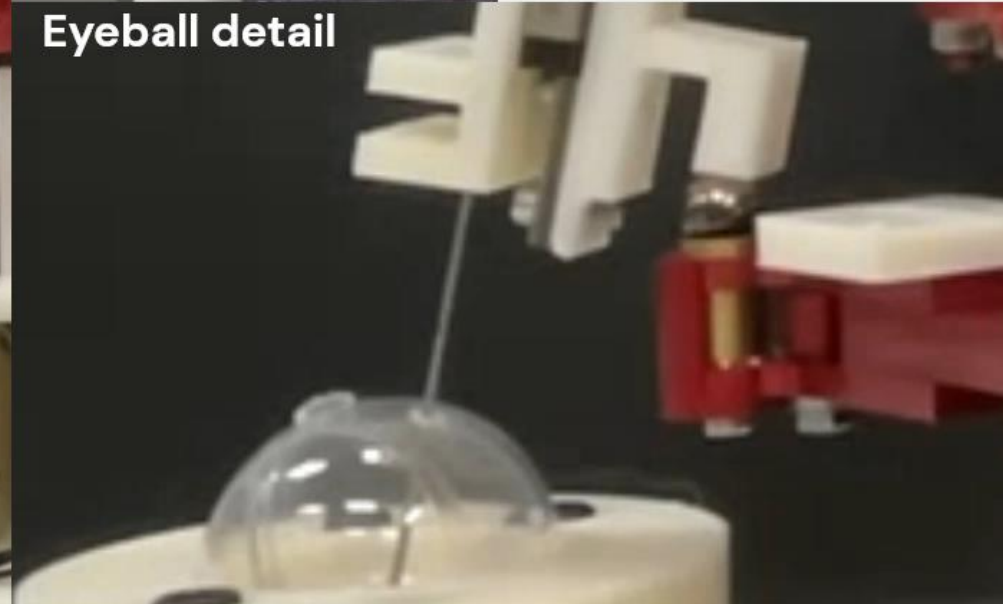
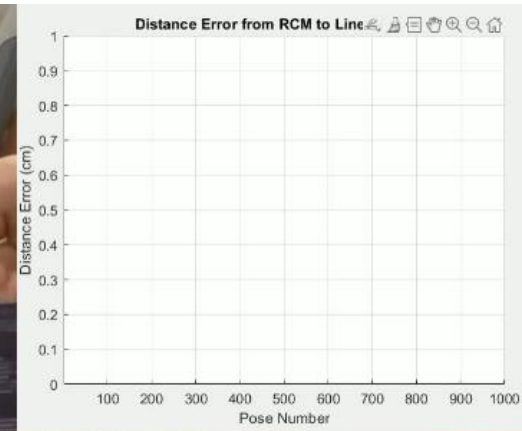
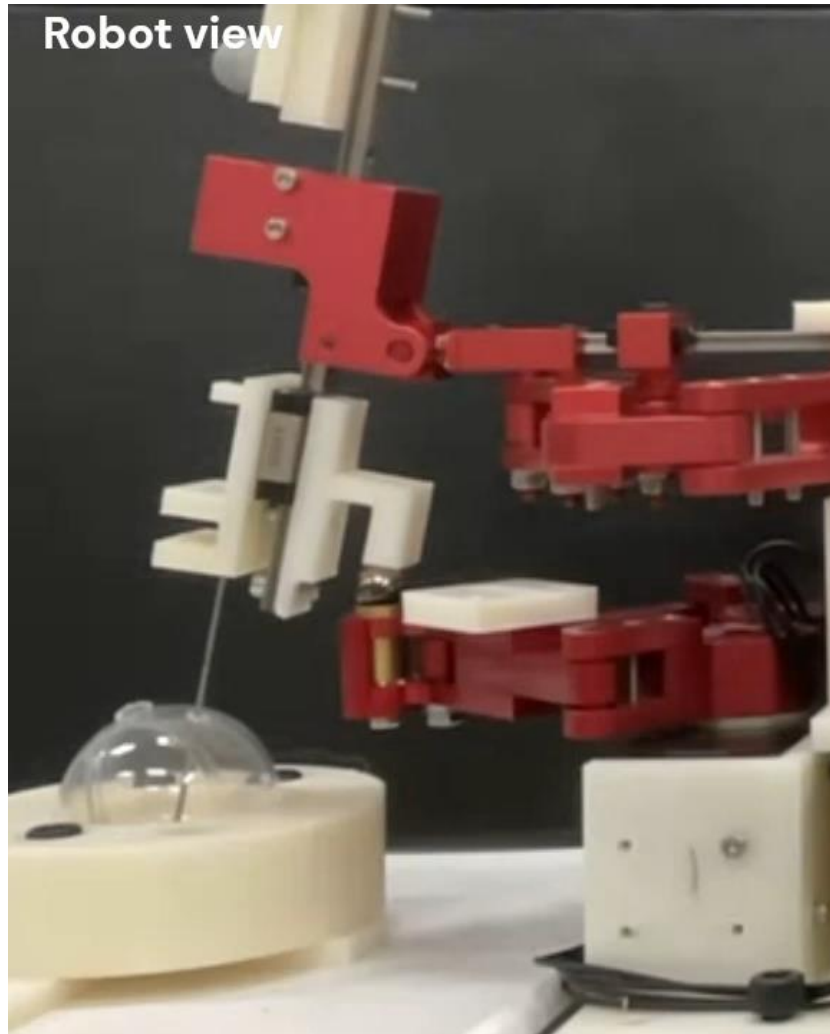
Block diagram of TSMCSPO parallel with parameter tuning of FIPSP using DDPG



Research



Research





Publications (59)

- ✓ 32 Journal Publication
- ✓ 27 Conference publication
- ✓ 1 Journal Paper (Under Review)
- ✓ Citation: 1151 (Google scholar)
- ✓ h-index: 16
- ✓ I10-index: 22

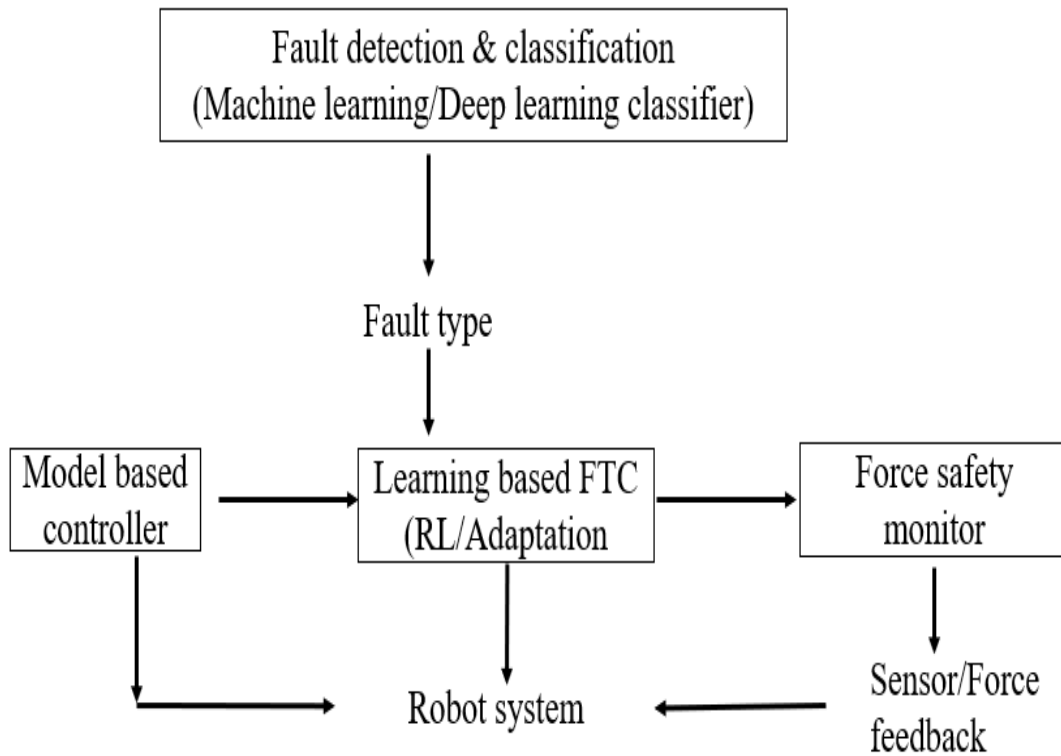


Future Research Plan

Hybrid Learning-Based Fault-Tolerant Control of Force-Sensitive Medical Robots

Potential Tools & Methods:

- a) Control: Impedance control, SMC, adaptive control, MPC
- b) Fault detection: Autoencoders, SVM, decision trees, LSTM
- c) Learning: Proximal Policy Optimization (PPO), DDPG, Q-learning
- d) Force Estimation: Observer-based force estimation (e.g., SPO), sensor fusion
- e) Simulation/Testing: ROS/Gazebo, MATLAB/Simulink, dVRK (da Vinci), or custom teleop system





Future Research Plan

In this research we integrate a novel framework which is an end-to-end large model wearable augmented vision device with a lower limb exoskeleton system for intelligent human-robot interaction and mobility support.

End-to-end large model wearable augmented vision device used as an observational and command interface.

End-to-end large model wearable augmented vision device analyzes real-time environmental circumstances, detects obstacles, and produces high-level semantic commands that are transmitted to a lower limb exoskeleton.

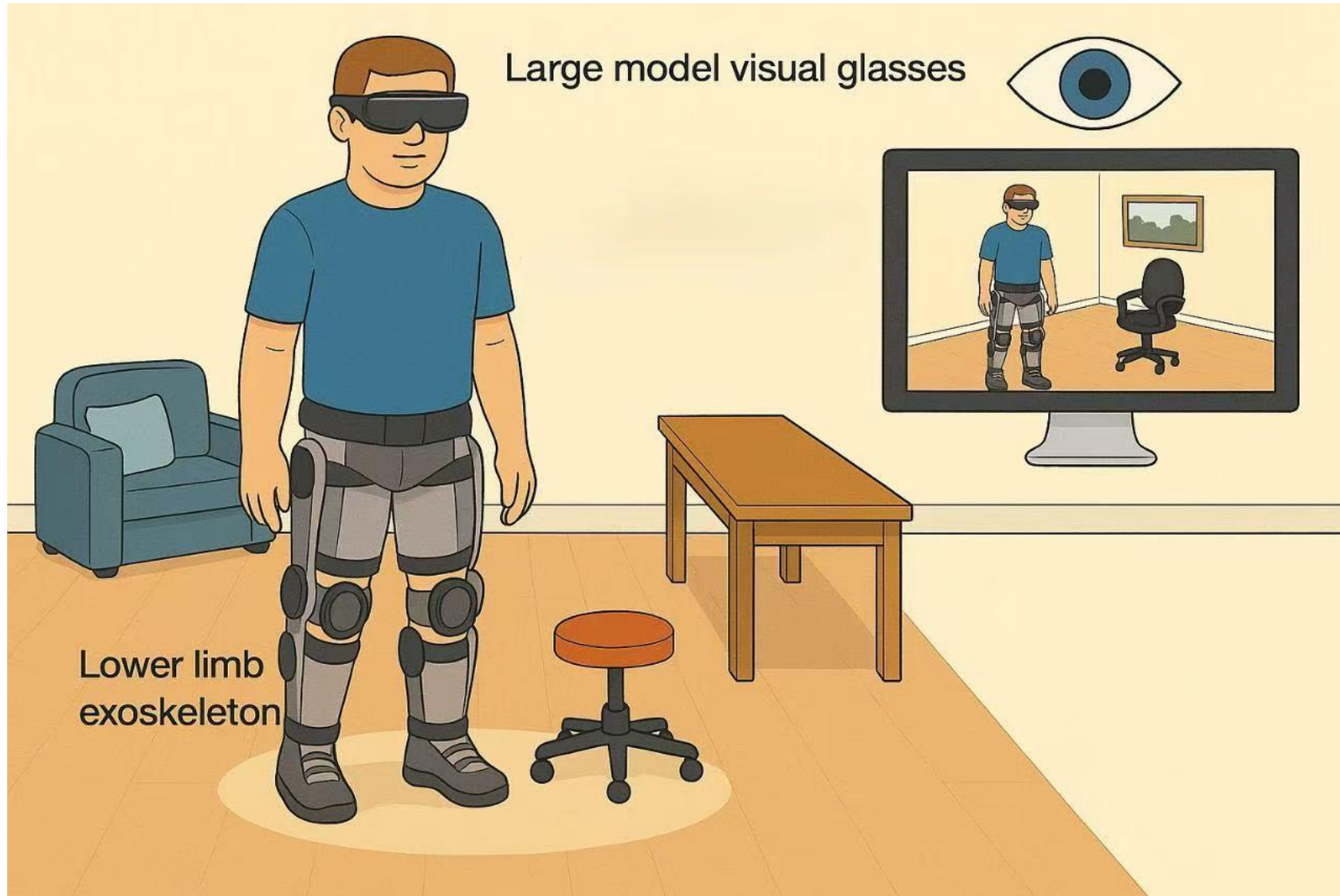
These commands are converted into coordinated motor actions using a biological rhythmic locomotion model based on Neuro-Inspired control.

The Neuro-inspired based controller supports adaptive gait generation and real-time modulation of walking patterns in response to both environmental signals and motion intention.

This biologically inspired control mechanism confirms smooth, stable, and adaptive movement, especially in unstructured or dynamic environments.

This research signifies an important step toward intelligent assistive mobility systems that combine cognitive scene understanding with biologically grounded motor control to improve mobility independence.

Future Research Plan





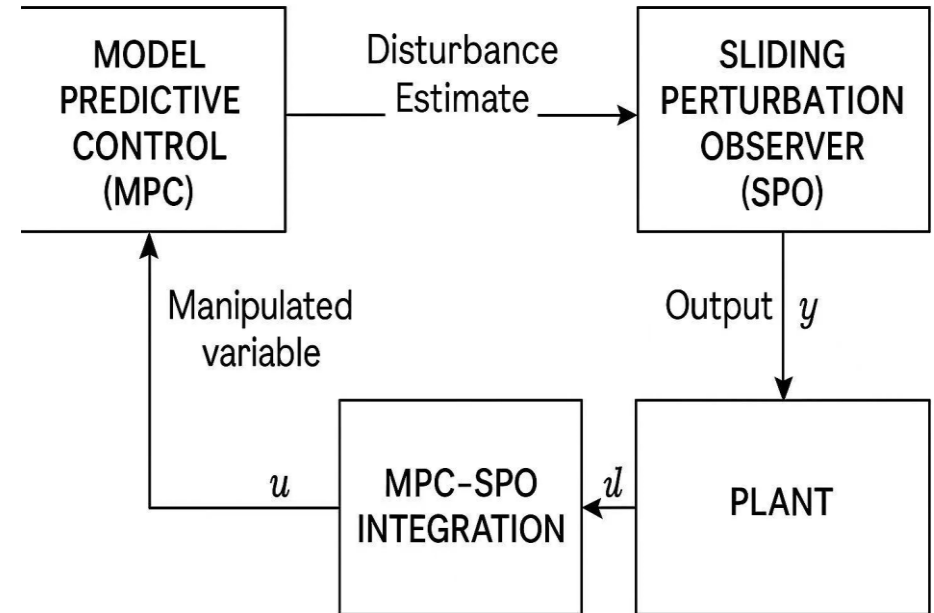
Future Research Plan

Title:

Integration of Model Predictive Control and Sliding Perturbation Observer for Robust Locomotion in Legged Humanoid Robots

Abstract:

This research proposes a novel control framework combining Model Predictive Control (MPC) with a Sliding Perturbation Observer (SPO) to achieve robust, adaptive, and predictive locomotion for legged humanoid robots. By leveraging MPC's anticipative control capabilities and SPO's robustness against model uncertainties and external disturbances, the integrated system aims to enhance dynamic stability during walking, running, and terrain transition





Future Research Plan

Introduction:

Humanoid robots often face significant challenges in maintaining balance and adaptability under real world disturbances and model inaccuracies. MPC offers optimal control with future state prediction, while SPO estimates and compensates for unknown perturbations. Their integration can result in a control scheme that is both predictive and resilient, particularly suitable for legged locomotion where dynamic stability is crucial.

Methodology:

The project will formulate a constrained MPC to plan optimal foot trajectories and control torques over a finite horizon. Simultaneously, an SPO will estimate external disturbances and compensate for them in real time. The combined controller will be tested in simulation and on a physical humanoid platform, focusing on disturbance rejection and terrain adaptability.

Expected Outcomes:

The integrated framework is expected to achieve improved disturbance rejection, smoother gait transitions, and enhanced adaptability to uneven terrains, outperforming traditional control approaches.



Course Contents

Introduction

- What is a robot?
- Classification of robots
- Robots in the Industry

The Course

- Spatial Description
- Manipulator Kinematics
- Jacobian
- Inverse Kinematics
- Dynamics of robots
- Path Planning and Trajectory Analysis
- Control



Books

- Text Book:
 - Introduction to Robotics, J. Craig, 3rd Edition, Prentice Hall



Grading Policy:

- **Lectures (3 credit hours):**

- 7x Quizzes (Best 6) = 10%
- 3x Assignments = 5%
- Paper Presentation= 10 %
- Project= 10 %
- Mid-term Exam = 25%
- Final Exam = 40%

* This grading policy is tentative

Any Questions?

